

1 Overview

This lecture surveys current Transformer research, focusing on efficiency, memory, and interpretability. Topics include linear Transformers, mixture-of-experts (MoE) architectures, and state-space models (SSMs) such as Mamba.

2 Linear Transformers

Linear Transformers (Katharopoulos et al., 2020) use a kernel trick to linearize the attention computation:

$$\text{LinearAttention}(Q, K, V) = \phi(Q)(\phi(K)^\top V)$$

where $\phi(\cdot)$ is a feature map applied elementwise. By computing $(\phi(K)^\top V)$ first, complexity drops from $\mathcal{O}(n^2)$ to $\mathcal{O}(n)$.

2.1 Common Kernels

Typical ϕ choices include:

- **ReLU kernel:** $\phi(x) = \text{ReLU}(x)$
- **ELU kernel:** $\phi(x) = \text{ELU}(x) + 1$
- **Softmax approximation:** $\phi(x) = \exp(x)/\sqrt{d_k}$

2.2 Tradeoffs

Linear Transformers offer strong computational and memory efficiency, especially for long sequences. However, scaled dot-product attention remains more expressive and empirically stronger on many benchmarks.

3 Other Transformer Variants

- **Sparse attention:** tokens attend to a subset of others (e.g., Longformer).
- **Low-rank approximations:** attention matrix projected into lower-dimensional space (e.g., Linformer).

4 Mixture of Experts (MoE)

MoE introduces sparsity in the MLP blocks of Transformers—only a subset of “experts” is active for each token. A gating network assigns mixing proportions p_k over experts:

$$E_c = \sum_k p_{ck} \exp\left(-\frac{\|t_c - o_{ck}\|^2}{2\sigma^2}\right)$$

Training increases weights of experts performing better than average, leading to specialization. Modern examples: **Mixtral 8×7B** and **Switch Transformer** (top-1 routing).

5 State-Space Models (SSMs)

SSMs model temporal dynamics explicitly:

$$x_t = Ax_{t-1} + Bu_t + w_t, \quad y_t = Cx_t + v_t$$

where x_t is a latent state and y_t the observation. They capture long-term structure efficiently and operate in continuous time.

5.1 Mamba (Gu & Dao, 2023)

Mamba combines SSM efficiency with attention-like adaptability:

$$h(t) = A(x_t)h(t-1) + B(x_t)x_t, \quad y(t) = C(x_t)h(t)$$

It computes transition parameters (A, B, C) as functions of input data, maintaining relevant context for long sequences.

6 Comparative Insights

- **Mamba advantages:** linear scaling, strong on long or streaming sequences, efficient for time-series.
- **Transformer advantages:** richer relational modeling, parallel training, multi-modal and transfer-learning strength.

7 Conclusion

Transformers remain dominant for general tasks, but linearized attention, MoE routing, and state-space formulations like Mamba are redefining efficiency and scalability for long-sequence modeling.